

Alview

AI-Based Interview Simulation and Evaluation System

Project Report

*Enhancing interview preparation through AI-driven
personalized feedback*

Shashwat Devan

**B.Tech in Computer Science and Communication Engineering
KIIT University**

Duration: Dec 2025 – Feb 2026

2. Abstract

Alview is an AI-based interview simulation and evaluation system designed to enhance interview preparation through personalized, automated feedback. Traditional interview preparation methods often lack real-time evaluation, structured feedback, and accessibility, making it difficult for candidates to assess and improve their performance effectively. This project addresses the problem by providing an interactive platform where users can upload their resumes, participate in AI-driven mock interviews, and receive detailed evaluations of their responses. The system leverages natural language processing and large language models to generate context-aware interview questions and evaluate user answers based on criteria such as technical accuracy, clarity, and relevance. Speech recognition and synthesis are integrated to simulate a realistic interview environment, allowing users to interact through voice as well as text. The platform follows a structured interview flow, progressing from background questions to technical and behavioral assessments, and presents results through a comprehensive dashboard with performance metrics and feedback. From a societal perspective, Alview democratizes access to high-quality interview preparation tools by providing scalable, personalized guidance without the need for human interviewers. It supports students and job seekers in improving their communication and technical skills, thereby enhancing employability and bridging the gap between academic learning and industry expectations.

3. Introduction

3.1 Background

Interview preparation plays a crucial role in shaping a candidate's ability to perform effectively in recruitment processes. Conventional preparation methods, such as practicing with static question lists or peer-based mock interviews, often lack structure and detailed feedback. This makes it difficult for candidates to assess their performance and improve systematically.

With the rise of Artificial Intelligence, recruitment processes have increasingly incorporated AI-based tools for screening, assessment, and evaluation. These systems are capable of analyzing responses, identifying patterns, and providing

automated feedback. As a result, there is a growing need for preparation platforms that simulate such AI-driven interview environments.

This project is developed as a prototype system to explore how AI can be used to enhance interview preparation through structured simulations and automated evaluation.

3.2 Problem Statement

Interview preparation often lacks a structured and feedback-driven approach, making it difficult for candidates to assess their performance effectively. While various resources are available, they do not always provide an integrated environment where users can practice interviews in a realistic and organized manner.

A key challenge is the absence of detailed and personalized feedback on user responses. Without clear insights into strengths and areas for improvement, candidates may find it difficult to refine their answers over time.

Additionally, interview practice is frequently conducted in a fragmented way, without a logical sequence or progression in question difficulty. This limits the ability to simulate real interview conditions and reduces the overall effectiveness of preparation.

3.3 Motivation

The motivation for this project originated from an interest in exploring how interview preparation can be enhanced through a more structured and interactive approach. The focus was on designing a system that integrates question generation, response evaluation, and feedback within a single workflow, allowing users to practice interviews in a more organized and consistent manner.

During the development process, it became evident that simulating an interview environment with a logical flow of questions and automated evaluation could provide a more effective way of practicing. This led to the idea of building a prototype system that combines question generation, answer evaluation, and performance feedback into a single workflow.

Another key consideration was that not all users are comfortable practicing interviews with other individuals. Factors such as hesitation, lack of confidence, or fear of being judged can reduce the frequency and quality of practice. A locally running AI-based system offers a private environment where users can attempt interviews multiple times without external pressure.

In addition to addressing these practical gaps, the project also serves as an exploration of integrating AI models into an interactive application. It provides insight into how AI-generated content and evaluation mechanisms can be combined to create a structured learning tool.

3.4 Scope of the Project

What the System Covers

- A prototype AI-based interview simulation system running on a local machine
- Generation of structured interview questions using AI models
- Evaluation of user responses with scoring and feedback
- A results interface for performance analysis
- A learning module with interview frameworks and example answers

What the System Does Not Cover

- Deployment as a publicly accessible or commercial platform
- Real-time interaction with human interviewers
- Fully accurate or bias-free AI evaluation (due to reliance on external AI models/APIs)
- Large-scale user handling, database management, or cloud-based infrastructure

4. Objectives

The primary objectives of the Alview project are as follows:

- To develop a prototype AI-based interview system that simulates a structured interview environment

- To generate personalized interview questions based on user-provided resume data
- To implement an AI-driven evaluation mechanism for analyzing user responses
- To provide structured and actionable feedback to help users improve their interview performance
- To explore the societal implications of AI-based evaluation systems, particularly in terms of accessibility, fairness, and user experience

5. System Overview and Architecture

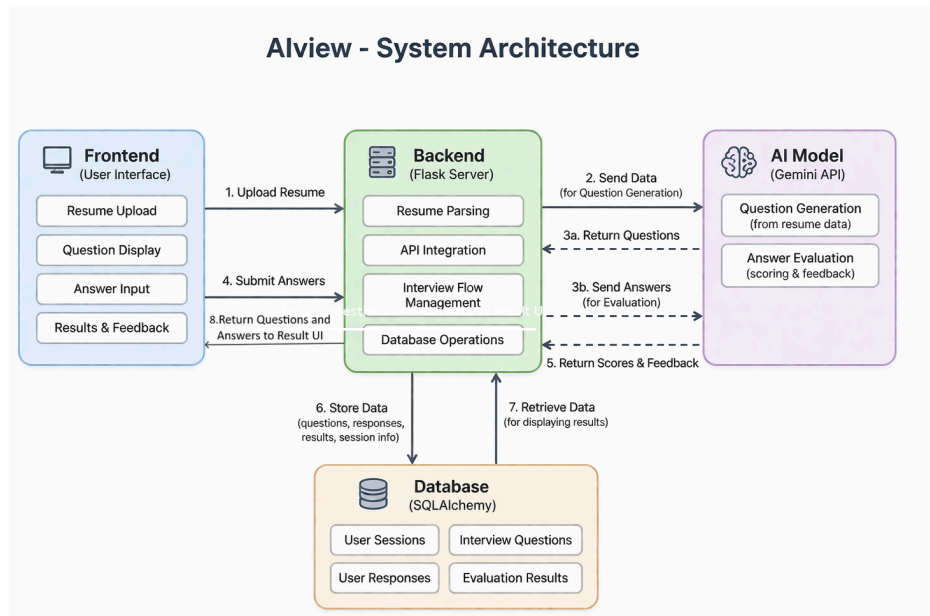
5.1 System Description

Alview is a prototype AI-based interview simulation system designed to provide a structured and interactive interview preparation experience. The system allows users to upload their resume, based on which personalized interview questions are generated using an AI model.

Users can respond to these questions, and their answers are evaluated automatically, providing scores and feedback. The system integrates multiple functionalities, including question generation, answer evaluation, and performance analysis, into a single workflow.

As a locally deployed prototype, Alview demonstrates the feasibility of combining AI-driven components with a user interface to simulate a realistic interview environment.

5.2 Architecture Diagram



5.3 System Components

1. Frontend (User Interface)

The frontend provides the interface through which users interact with the system. It is responsible for resume upload, displaying interview questions, accepting user responses, and presenting evaluation results. The interface is designed to simulate a structured interview environment.

2. Backend (Flask Server)

The backend, implemented using Flask, acts as the core processing unit. It handles user requests, processes resume data, communicates with the AI model, manages interview flow, and interacts with the database.

3. AI Model (Gemini API)

The AI model generates interview questions and evaluates user responses. It produces structured outputs including scores and feedback. The system depends on external API calls, which may be subject to availability and latency constraints.

4. Database (SQLAlchemy)

The database, managed using SQLAlchemy, stores interview data such as questions, user responses, evaluation results, and session-related information. This enables result tracking and performance visualization.

5.4 Data Flow

The system follows a sequential workflow:

1. Resume Upload

The user uploads their resume through the frontend interface.

2. Resume Parsing

The backend processes the resume to extract relevant information such as skills, education, and experience.

3. Question Generation

The extracted data is processed by the AI model to generate a structured set of interview questions.

4. Interview Process

Questions are presented to the user sequentially, and responses are recorded.

5. Answer Evaluation

The system presents questions to the AI model for evaluation, generating scores and feedback.

6. Data Storage

All interview data, including questions, answers, and evaluations, are stored in the database.

7. Result Display

The system displays a summary of performance along with detailed feedback through the frontend dashboard.

6. Technical Implementation

6.1 Technology Stack

The Alview system is built using a combination of frontend, backend, and AI technologies:

- Python and Flask are used for backend development and handling application logic
- HTML, CSS, and JavaScript are used to design and implement the user interface
- Gemini API is used for generating interview questions and evaluating user responses
- SQLAlchemy is used for database management and ORM functionality
- pdfplumber and python-docx are used for extracting text from PDF and DOCX resumes

6.2 Resume Parsing Module

The resume parsing module is responsible for extracting relevant information from user-uploaded resumes.

- Supports PDF and DOCX file formats
- Uses pdfplumber and python-docx for text extraction
- Performs basic text cleaning to remove unnecessary formatting and noise
- Extracted content is passed to the backend for further processing and question generation

6.3 Question Generation Module

This module generates interview questions based on the parsed resume data.

- Uses prompt engineering techniques to guide the AI model
- Generates structured output in JSON format for consistency
- Ensures a logical flow of questions (education → projects → skills → behavioral)
- Incorporates difficulty levels (easy, medium, hard) for progressive assessment

6.4 Interview Engine

The interview engine manages the interaction between the user and the system.

- Displays questions sequentially through the UI
- Accepts user responses via text and speech input
- Maintains interview state and flow
- Includes a timer-based response system to simulate real interview conditions

6.5 Answer Evaluation Module

This module evaluates user responses using the AI model.

- Evaluates responses using the Gemini API
- Assigns scores on a 0–10 scale
- Generates structured feedback including strengths, weaknesses, reasoning, and suggestions
- Uses a weighted scoring mechanism based on question difficulty

6.6 Database Design

The database stores all interview-related data and maintains relationships between entities.

- **Interview Table:**
Stores session-level data such as interview ID, timestamp, and overall score
- **Answer Table:**
Stores individual question responses, scores, and feedback
- **Relationship Structure:**
One-to-many relationship between interview sessions and answers

6.7 Challenges and Solutions

During development, several challenges were encountered:

- **API Response Inconsistencies:**
AI-generated responses were sometimes unstructured; addressed using stricter prompt design and validation checks
- **JSON Parsing Issues:**
Occasional malformed outputs were handled using error handling and fallback parsing mechanisms
- **Handling Missing Inputs:**
Edge cases such as empty responses or incomplete resumes were managed through input validation
- **Performance and Latency:**
API response delays were mitigated by optimizing request handling and limiting unnecessary calls

7. Societal and Ethical Analysis

7.1 Impact on Interview Preparation

The Alview system contributes to improving accessibility in interview preparation by providing a structured and self-paced practice environment. Users can simulate interviews without relying on external resources or scheduling constraints.

It also supports the democratization of interview practice by making guided preparation available to a wider range of users, regardless of their access to professional coaching or peer networks.

7.2 Bias and Fairness

AI-based evaluation systems may introduce bias depending on the underlying model and training data. Since the system relies on an external AI model (Gemini API), the evaluation may reflect inherent biases present in the model.

There is also a possibility that responses are judged based on linguistic style or phrasing rather than actual content quality, which can affect fairness across different users.

7.3 Privacy Concerns

The system processes user resumes, which may contain sensitive personal information such as education, experience, and contact details. Proper handling of this data is important to ensure user privacy.

Although the current implementation operates locally, data is still stored within the system database. Therefore, considerations around secure storage and restricted access remain relevant.

7.4 Reliability of AI Systems

The system depends on AI-generated outputs for both question generation and answer evaluation. Due to the probabilistic nature of large language models, outputs may vary across different runs.

This variability can lead to inconsistencies in scoring and feedback, and there is a risk of occasional incorrect or less relevant evaluations. Hence, AI outputs should be interpreted as supportive guidance rather than absolute judgment.

7.5 Accessibility Considerations

The system is designed with a user-friendly interface to ensure ease of use for a broad audience. Features such as structured layouts, clear feedback, and guided interaction improve usability.

However, factors such as language proficiency and communication style may influence user performance, especially in AI-based evaluation. Future improvements can focus on supporting diverse communication styles and multiple languages.

8. Results and Evaluation

8.1 System Output Examples

The Alview system produces structured outputs for both question generation and answer evaluation.

- **Sample Generated Questions:**

The system generates personalized interview questions based on resume content, covering areas such as education, projects, skills, and behavioral aspects. Questions follow a logical sequence and include varying difficulty levels.

- **Sample Evaluated Answers:**

User responses are evaluated by the AI model, producing:

- A score on a scale of 0–10
- Strengths of the response
- Weaknesses or missing aspects
- Reasoning behind the evaluation
- Suggestions for improvement

These outputs demonstrate the system's ability to simulate a realistic interview and provide actionable feedback.

8.2 Performance Evaluation

- **Accuracy (Qualitative):**

The system provides reasonably consistent evaluations of user responses. The feedback aligns with general expectations of interview performance, although minor variations may occur due to the probabilistic nature of the AI model.

- **User Experience:**

The interface enables smooth interaction, with clear question display, response input, and result visualization. Features such as structured feedback, scoring, and performance tracking enhance usability and engagement.

8.3 Observations

- **Strengths of the System:**

- Integrated workflow from resume upload to final evaluation
- Personalized question generation
- Structured and detailed feedback
- Performance tracking and interview history
- User-friendly interface design

- **Real-World Usability:**

The system demonstrates practical usability as a standalone interview preparation tool. It can assist users in improving their responses through repeated practice and feedback. However, it is intended as a supportive system rather than a replacement for real interview experience.

9. Limitations

The Alview system, while functional and effective as a prototype, has certain limitations:

- **Dependence on Gemini API:**

The system relies on the Gemini API for question generation and answer evaluation. Any issues such as API downtime, rate limits, or latency can directly affect system performance.

- **Lack of Real Human Judgment:**

The evaluation is entirely AI-driven and does not incorporate human judgment. As a result, certain nuanced aspects of communication, such as tone, intent, or context, may not be fully captured.

- **Possible Scoring Inconsistencies:**

Due to the probabilistic nature of AI models, similar responses may receive slightly different scores or feedback across multiple evaluations.

- **Limited Contextual Understanding:**

While the system analyzes responses based on provided input, it may not fully understand deeper context, domain-specific nuances, or implicit meanings in answers.

10. Future Work

The Alview system can be further enhanced in several directions to improve functionality, scalability, and real-world applicability:

- **Improved Evaluation Models:**

Future versions can incorporate more advanced or fine-tuned AI models to improve the accuracy and consistency of answer evaluation, especially for domain-specific questions.

- **Offline / Local AI Models:**

Integrating locally hosted AI models can reduce dependence on external APIs, improve privacy, and enable the system to function without internet connectivity.

- **Enhanced UI/UX:**

The user interface can be further improved with more interactive elements, better visualization of performance analytics, and smoother navigation to enhance overall user experience.

- **Multi-Language Support:**

Supporting multiple languages can make the system accessible to a broader audience and allow users to practice interviews in their preferred language.

- **Integration with Real Hiring Systems:**

The system can be extended to integrate with real recruitment platforms, enabling its use in preliminary screening or candidate assessment workflows.

11. Conclusion

The Alview project successfully demonstrates the design and implementation of a prototype AI-based interview simulation system. The system integrates resume-based question generation, AI-driven answer evaluation, structured feedback, and performance tracking into a single workflow. This provides users with a practical and interactive platform for improving their interview skills.

From a technical perspective, the project showcases the effective use of technologies such as Flask, SQLAlchemy, and the Gemini API to build an end-to-end system. It highlights how AI models can be integrated into user-facing applications to perform complex tasks such as natural language evaluation and personalized content generation.

From a societal perspective, the system contributes toward making interview preparation more accessible and flexible. By providing a private and structured environment, it enables users to practice without external pressure and receive feedback for continuous improvement. At the same time, the project acknowledges the limitations and ethical considerations associated with AI-based evaluation systems.

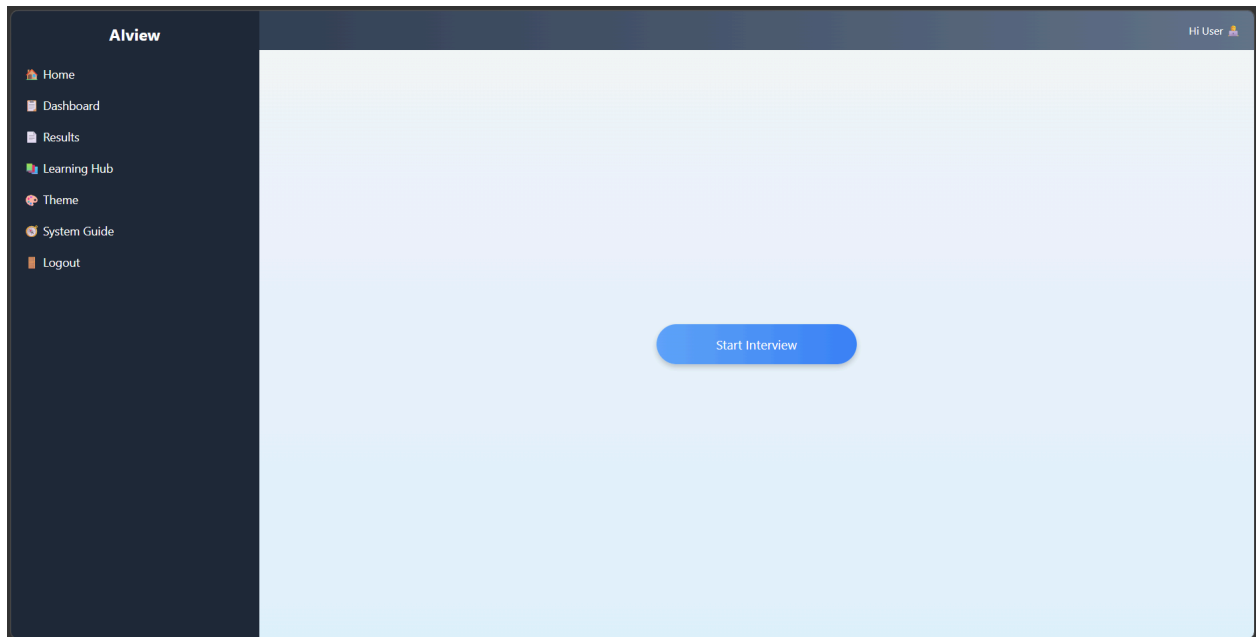
Overall, Alview serves as a proof-of-concept that demonstrates the potential of AI in enhancing interview preparation while also identifying areas for future improvement and real-world applicability.

12. References

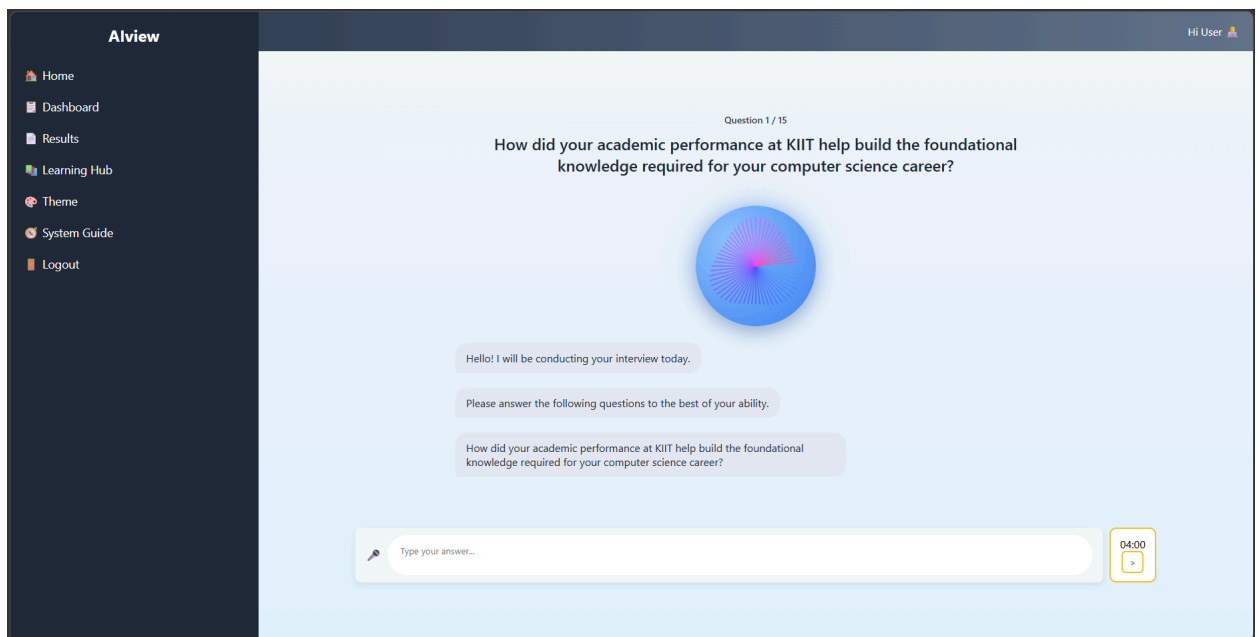
- Google AI, "Gemini API Documentation," Available: <https://ai.google.dev>
- Pallets Projects, "Flask Documentation," Available: <https://flask.palletsprojects.com>
- pdfplumber Documentation, Available: <https://github.com/jsvine/pdfplumber>
- python-docx Documentation, Available: <https://python-docx.readthedocs.io>
- OpenAI, "ChatGPT (GPT-based AI Assistant)," Used for UI design guidance and development support, Available: <https://chat.openai.com>

13. Appendix

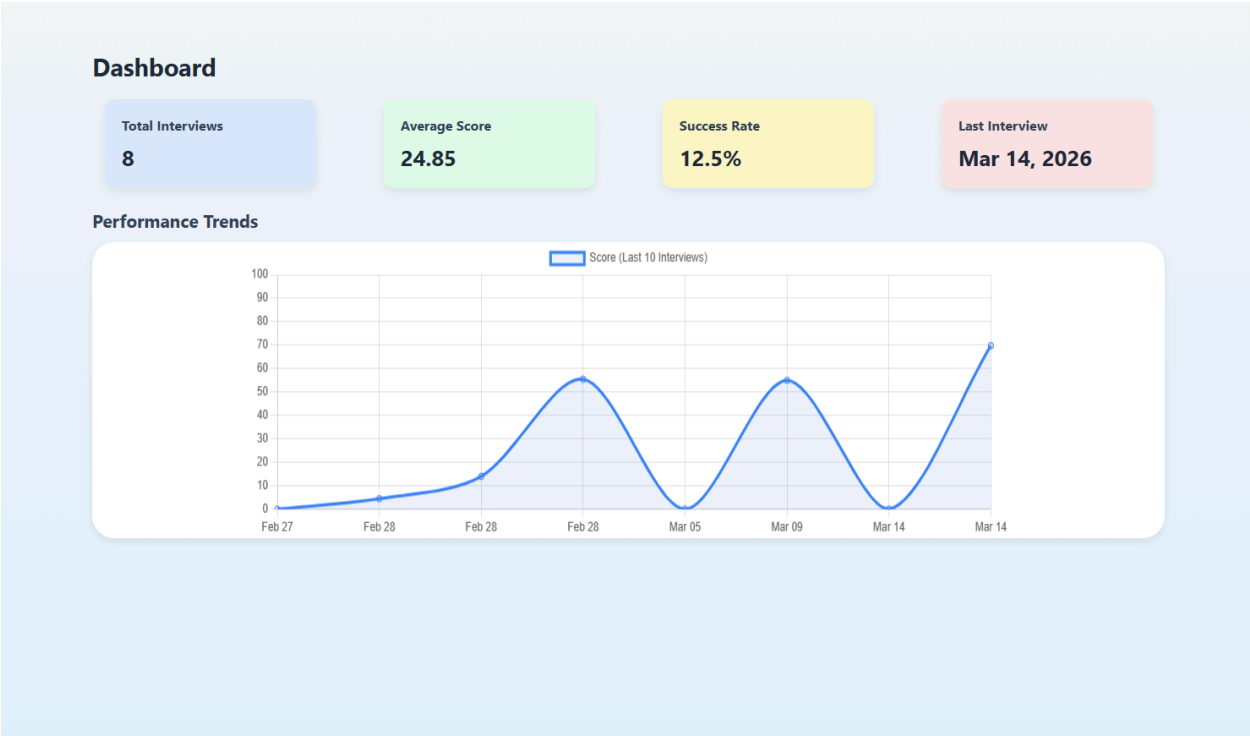
- Screenshots of UI:
 - Home page:



- Interview page:



➤ Dashboard page:



➤ Results page:

Interview History

Date	ID	Score	Performance	Details
15-03-2026, 00:15	8	69.8/100	Good	View
14-03-2026, 13:16	7	0.0/100	Below Average	View
09-03-2026, 11:57	6	55.0/100	Fair	View
05-03-2026, 10:59	5	0.0/100	Below Average	View
28-02-2026, 20:33	4	55.5/100	Fair	View
28-02-2026, 10:22	3	14.0/100	Below Average	View
28-02-2026, 07:48	2	4.5/100	Below Average	View
27-02-2026, 17:03	1	0.0/100	Below Average	View

➤ Learning hub Page:

[Interview Tips](#) [Resume Tips](#) [Common Questions](#) [Answer Frameworks](#) [AI Answer Examples](#)

Interview Tips

- 1. Research the Company**
Understand the company's mission, products, culture, and recent news. This helps you tailor your answers.
- 2. Practice Structured Answers**
Use frameworks like STAR to give clear and concise responses.
- 3. Communicate Clearly**
Speak slowly, organize your thoughts, and avoid unnecessary filler words.
- 4. Highlight Real Experiences**
Employers value practical examples from projects, internships, or coursework.
- 5. Prepare Questions**
Always ask thoughtful questions about the role, team, or growth opportunities.

➤ System Guide Page:

 **System Guide**
This guide explains how to use Alview effectively and provides tips to ensure the best interview experience.

Preparation Tips

- **Use Microsoft Edge** for the best speech recognition and audio experience.
- **Wear earphones or headphones** to prevent the AI voice from being picked up by the microphone.
- **Login before starting** to access Dashboard, Results, and Learning Hub features.
- **Choose a quiet environment** to improve speech recognition accuracy.
- **Speak clearly and at a moderate pace** when answering questions.

How to Use Alview

Step 1 — Login

Click the Login button in the top right corner and enter your credentials.

